

Spark Plug Problem

Claim

Average Lifetime = 28000

Hypotheses

$H_0: \mu = 28000$

$H_A: \mu < 28000$

```
xbar <- 26040 # sample mean
sigma <- 12894 # population stdev
n <- 27 # sample size
mu.0 <- 28000 # null value
stderr <- sigma / sqrt(n)
alpha <- 0.05 # significance level
```

Analyze Data

Find z-score

```
z <- (xbar - mu.0) / stderr
```

```
pnorm(z)
```

```
## [1] 0.2148047
```

Web Browser Hypothesis Test

```
p <- 0.99
```

```
n <- 1014
phat <- 984 / n
se <- sqrt(phat * (1 - phat) / n)
alpha <- 0.05
```

```
zstar <- qnorm(1 - alpha / 2) # Critical Value
me <- se * zstar
conf_itv <- c(phat - me, phat + me)
```